

COPELAND ROYALL

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copey.dev

SKILLS

Languages Python, TypeScript, Go

Software AWS, GCP, Terraform, Terragrunt, Pulumi, Ansible, GitHub Actions, Docker, Kubernetes

EXPERIENCE

Freelance Software Engineer

January 2025 – present

Opinion Health

Architected and built a pharmaceutical research AWS environment using Terraform and Terragrunt

- Designed account-isolated RDS Aurora architecture to enforce per-region data residency for healthcare data
- Isolated org CloudTrail and pgAudit database logs in a dedicated log-archive account, exporting via Lambda to meet 6-year compliance retention
- Enforced org-wide guardrails via SCPs (region locks, mandatory encryption, KMS deletion protection)
- Delegated GuardDuty/CloudTrail administration to a dedicated security account
- Built a GitOps CI/CD pipeline in GitHub Actions with Terraform module unit tests and security/secret scanning, gating every deploy on PR review and automated drift detection
- Built a Python NLP pipeline for extreme multi-label classification (1.4k labels) across 130k patient records spanning 10 years, combining traditional techniques with Bedrock embeddings and batch LLM inference; anonymized patients until database ingest to minimize exposure risk
- Restricted database access to SSO/MFA-enforced sessions from Ansible-provisioned VMs
- Eliminated Aurora cold-start query timeouts by triggering a warmup Lambda on user VM login

Stealth Startup

- Designed self-hosted Kubernetes platform on declarative NixOS hosts, GitOps-managed by Flux CD
- Built cross-repo GitHub Actions release pipelines; staging auto-promoted via Flux image automation
- Deployed Helm kube-prometheus-stack, Loki/Fluent Bit logging, self-hosted error tracking, SOPS secrets

Keygraph

- Developed multi-account, -region and -environment AWS platform module in TypeScript/Node.js Pulumi IaC
- Deployed 6 containerized production Go apps using App Runner, RDS Aurora and SOPS secrets

Historical Artist Website

- Built website for acclaimed artist with Hugo, Cloudflare Worker and R2; JavaScript OAuth proxy for editing

Visa Software Engineer

September 2023 – May 2024

Accelerated EU payment product rollouts on GCP with TypeScript IaC and led deployment of cost monitoring in org.

- Implemented TypeScript Pulumi IaC package deploying 256 resources across 4 GCP projects, meeting PCI compliance requirements
- Wrote reusable Pulumi packages for GKE, IAM, and KMS resource management with comprehensive TypeScript unit test coverage
- Led team initiative to deploy Cloudability FinOps in 45 test and production projects; calculated 10% cost savings
- Installed and updated Kubernetes cluster addons using Argo CD, Kustomize and Helm
- Migrated Concourse CI pipelines to new load-test platform

Visa DevOps Engineering Intern

Summer 2022

Developed foundational Terraform modules for AWS RDS Aurora and ElastiCache Redis including best-practice defaults to enforce consistency across teams.

Tangible AI Natural Language Processing Intern

Summer 2021

EDUCATION

University of Edinburgh BSc Computer Science, First-class honors

2019 – 2023

Dissertation: Microservices and message queue-based multi-receiver ADS-B aircraft tracking data aggregation system written in Go, utilizing Apache Kafka, MongoDB and GraphQL, achieving 10ms end-to-end latency

PROJECTS

- Built first 3D aircraft flight heatmap and radar viewer to see paths curving with the globe (instead of distorted 2D)
- Empowered Siri with local aircraft tracking, achieving over 18,500 downloads from GHCR